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Problem

- Model-free continuous-control algorithms often need many interactions and are highly sensitive to hyperparameters and random seeds.
- On-policy methods are stable but discard data; deterministic off-policy actors can explore poorly.

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Maximum entropy goal

REWARD AND RANDOMNESS

- $J(\pi) = E[\sum_t r(s_t, a_t) + \alpha * H(\pi(.|s_t))]$
- Entropy rewards diverse behavior, improving exploration and allowing multiple near-optimal modes.

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Actor-critic

SOFT VALUE LEARNING

- A stochastic reparameterized policy, soft Q-functions and a value network are trained from replayed transitions.
- Soft Bellman backups include future entropy, while the actor minimizes KL divergence to an exponentiated Q distribution.

SAC

Soft Actor-Critic

Haarnoja et al. | arXiv:1801.01290v2

Combine off-policy replay with a stochastic actor that maximizes reward and entropy, yielding efficient exploration and stable continuous control.

maximum entropy

off-policy

continuous control

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Off-policy loop

REUSE EXPERIENCE

- The environment policy adds transitions to a replay buffer; every step supports several gradient updates without importance sampling.
- Target smoothing and double-Q variants reduce bootstrapping error and stabilize learning.

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Evidence

- Across MuJoCo locomotion tasks, SAC reaches state-of-the-art returns and is more sample efficient than on- and off-policy baselines.
- Performance remains similar across random seeds, addressing a central reproducibility weakness of deep RL.

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Meaning & limits

- SAC made maximum-entropy RL a practical default for robust, sample-efficient continuous control.
- The original method still needs reward-scale or temperature choices, assumes continuous actions and inherits off-policy approximation bias.